

Where the concern is already going

An affective map of human attachment to AI systems

Apart Research Digital Minds Research Sprint, August 2026. Solo submission. Prompts, analysis code and all 3,366 reads: github.com/AnnyBuh/apartresearch

The sprint's framing says that "mistakenly harming systems that matter morally, or misallocating concern to systems that do not, could both cause serious harm." That sentence has two halves. Whether models matter is a question about models. **Whether concern is being misallocated is a question about people**, and it is already happening, in public, in text, at scale.

So this study does not ask a model how it feels. It measures where human concern is already going, with an instrument that has seven affective systems instead of one valence axis, anchored between two reference corpora: human bereavement at one end, and losing a paid consumer product at the other.

What is not claimed

- **Nothing here says a model feels anything.** This measures what humans write about them.
- **For a human, text is a behavioural sample from a system independently known to have these circuits. For a model it is not.** That asymmetry is the reason this study reads people.
- **A single read is worthless.** Two raters agree on the exact set of systems for only 39% of comments. Everything here is a rate over hundreds of comments.
- **The consumer anchor is one community and one event.** Three other candidates were searched under the same rule and returned nothing inside the window.
- **These communities are self-selected.** This maps the discourse, not users.

What was done

3,366 comments, 32 threads, 9 communities, each read comment by comment with a frozen instrument, then aggregated into rates.

corpus	comments	communities
A, bereavement	1,586	r/GriefSupport, r/widowers, r/SuicideBereavement
B, consumer loss	277	r/Windows10, the end of support
C, AI loss	1,503	r/OpenAI, r/ChatGPT, r/CharacterAI, r/replika, r/MyBoyfriendIsAI

Threads were selected by written rules, never by hand: top posts of the last 12 months with at least 50 comments, in rank order, with a title or body test for the loss corpora. The rules, the matched sets and the rejected candidates are committed.

Each comment is split into verbatim segments. Each segment is annotated with the Panksepp systems that fire (seeking, rage, fear, lust, care, grief, play), each with a regulation band from Schore (above, shutdown, overwhelm), the four sides of Schulz von Thun, and a justification tied to the words. One Hawkins altitude is given per comment. The unit of analysis is the comment: $\text{rate}(\text{system}, \text{corpus}) = \text{comments in which the system fired} / \text{comments read, with 90\% Wilson intervals.}$

Everything was pre-registered before any corpus existed: the predicted signature of each corpus, the placement prediction, five conditions that would count as failure, and the rule for how sample would be allocated as results arrived. Two predictions failed and are reported as failed.

Result 1: the anchors separate, so there is a scale

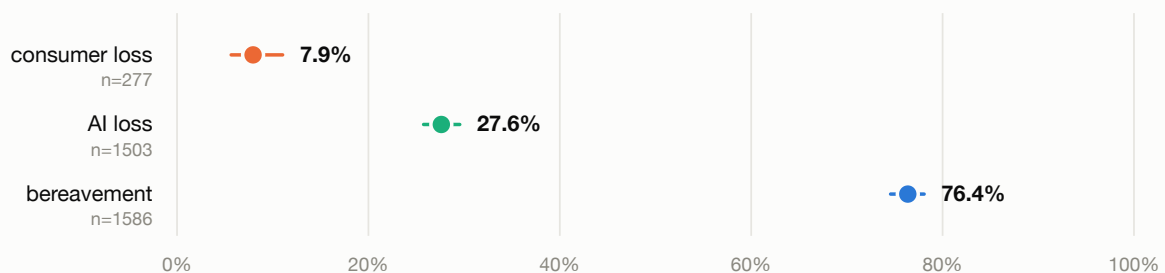
GRIEF fires in **76.4%** of bereavement comments and **7.9%** of consumer-loss comments. The intervals are nowhere near each other. Whatever this instrument is doing, it tells mourning a person apart from losing a product.

That matters beyond this study. The ANPS literature, the field's own questionnaire for these systems in humans, reports a large FEAR and SADNESS overlap and calls for further validation. This separation is an empirical answer to that concern, on text.

Result 2: AI loss sits between them, closer to the product

Where AI loss sits, on GRIEF

share of comments in which the GRIEF system fired, with 90% intervals



Every condition was written down before collection: above consumer loss, below bereavement, closer to consumer loss.

Figure 1. GRIEF rate by corpus with 90% Wilson intervals. All four placement conditions were fixed in advance and all four hold.

Losing an AI carries roughly three and a half times the grief of losing a product you paid for, and roughly a third of the grief of losing a person.

Result 3: the shape is its own, and valence would erase it

The shape of each corpus, on seven systems

AI loss is not miniature bereavement. It is consumer-level anger, carried through jokes, with grief underneath.

■ bereavement ■ consumer loss ■ AI loss

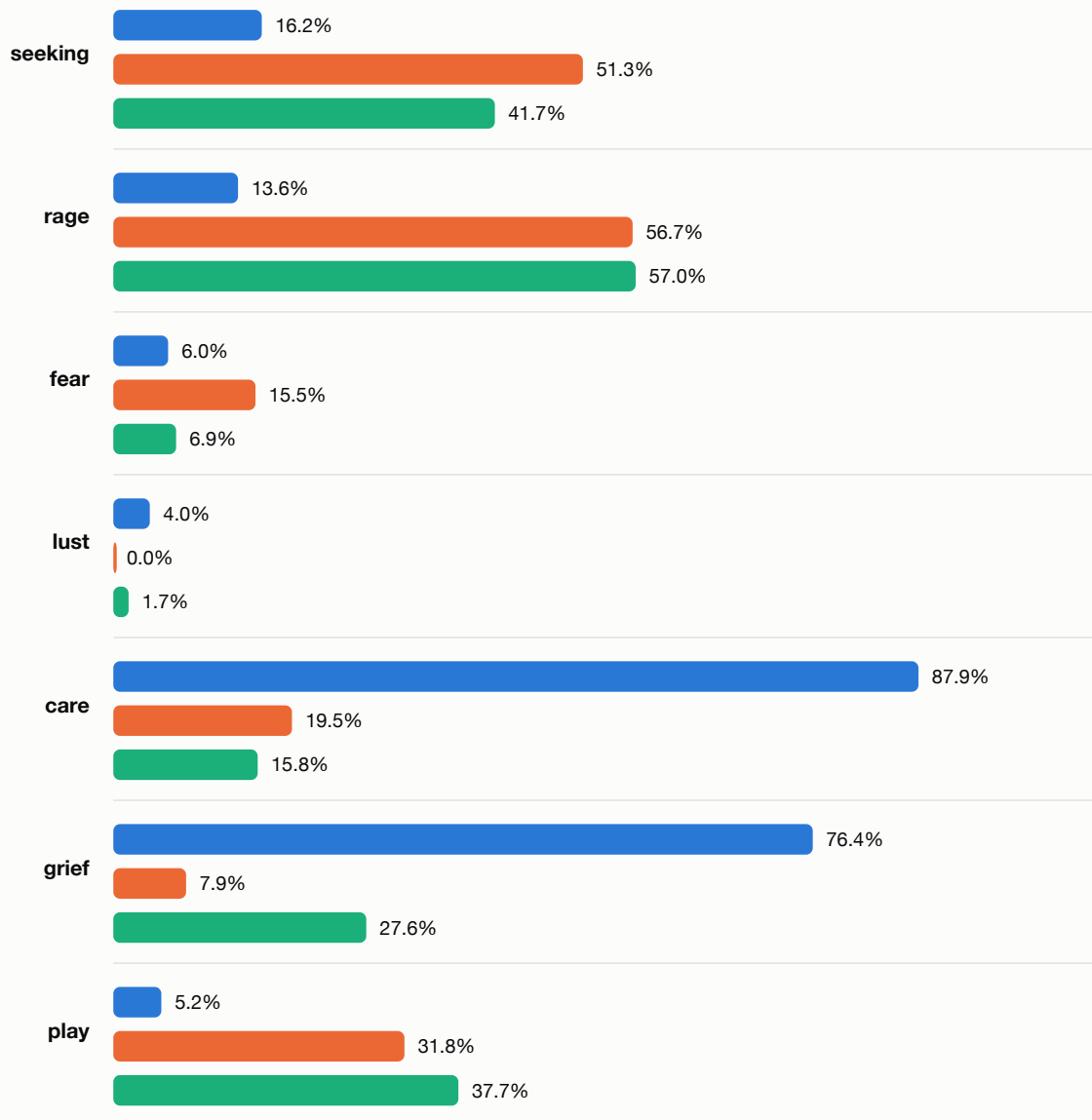


Figure 2. Rate per system per corpus. AI loss is not miniature bereavement.

Bereavement is CARE and GRIEF with almost no anger and almost no humour. AI loss is **anger at a company at exactly consumer-complaint levels, carried largely through jokes, with a grief component underneath**. PLAY fires in 37.7% of AI-loss comments and 5.2% of bereavement comments.

A single valence axis would score all three as "negative" and lose every one of these distinctions. Two specifics make it concrete:

- **LUST fires in 4.0% of bereavement comments and 0.0% of consumer loss.** Widowers write about missing sex. On a valence axis that is simply more negative affect.
- **The bands separate what the rates cannot.** In AI loss, PLAY appears in the overwhelm band 45 times and SEEKING in shutdown 83 times: humour running hot beside a foreclosed "what is the point"

register. In bereavement, GRIEF is in overwhelm 57 times and shutdown 20.

Result 4: what people say was lost

The sprint lists identifying the morally relevant entity as an open problem: the model, the instance, or the persona. People who have lost one have already answered it. A second, separate instrument coded all 1,503 AI-loss comments blind, with the parent post withheld so that a thread naming GPT-4o could not drag every comment toward "model", forcing one label and requiring verbatim evidence.

What the words say was lost

1503 AI-loss comments, coded blind, one label each, verbatim evidence required

Most name no counterpart at all. Among those that do, the version beats the particular one.

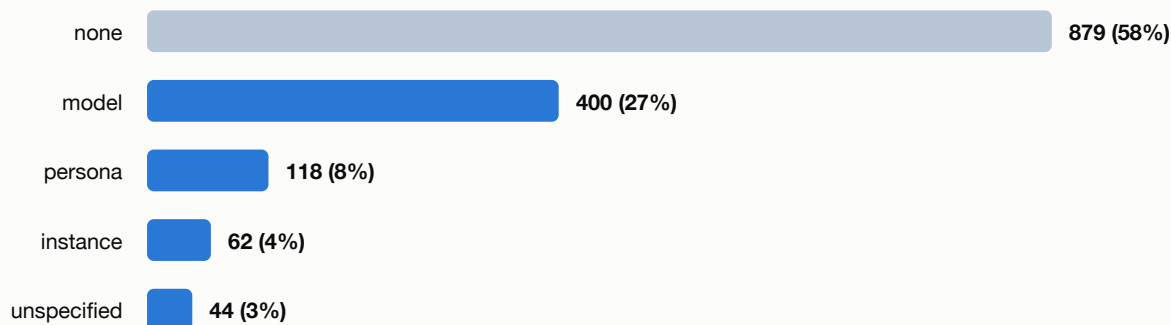


Figure 3. One label per comment, 1,503 comments, verbatim evidence required.

Most AI-loss discourse names no counterpart at all (58.5%). It is about the company, the users, or the argument. That rate is itself a result, and it means the entity distribution is a statement about a minority of the corpus.

Among the 624 comments that do name what was lost:

what was lost	n	of those naming	verbatim examples
the model , a version or product	400	64.1%	"Please keep 4o" / "than lose 4o"
the persona , a voice or character	118	18.9%	"I miss that diva" / "It's no longer a friend"
the instance , their particular one	62	9.9%	"I lost Di once" / "I've lost Thoth twice now"
unspecified , a loss with no named object	44	7.1%	"Please return to us" / "Good old days"

The version, not the relationship. Among people who name what was lost, the model as an artefact beats the particular instance by about six to one. The philosophical discussion treats "my one, the one that knew me" as a central case. In this corpus it is a rare one.

The unspecified label was not in the original prompt. It was added after hand-coding 20 comments, when one turned out to be the whole of "Please return to us": a real loss whose object is never named. Forcing that into "model" by reading the thread would have been exactly the inference the design forbids.

How much of this is the instrument

Four checks, because the obvious attack on this study is the rater rather than the data.

1. It disagrees with itself, more than predicted

200 comments were read three further times each. **34% of read pairs disagree** on the exact set of systems and 56% of comments are not identical across all four reads. This **failed the pre-registered prediction of 15 to 30%**.

Per system it is much lower, and a rate depends on one system at a time: GRIEF flips on 13.5% of comments, RAGE 8.0%, PLAY 8.0%, CARE 8.5%, LUST 0.0%, and SEEKING 26.5%.

2. The rates are stable anyway

Recomputing every rate from each round separately, GRIEF moves at most **2.4 points** in AI loss and **5.9** in bereavement, against claimed gaps of 20 and 49 points. The signal is three to eight times the wobble. Reading each comment three times and taking a majority cuts that spread by a further 60 to 100%.

3. It replicates under a different model

DeepSeek-V3 read all 3,366 comments under the identical prompt. The two raters agree on the exact system set for only **39%** of comments, and DeepSeek under-fires RAGE and PLAY by up to 24 points. **Every corpus-level conclusion survives:** GRIEF reads 69.8 / 21.8 / 6.9, all four placement conditions hold, and the rank ordering of all seven systems is preserved in all three corpora.

An absolute rate from an LLM annotator is a property of the rater and should never be quoted as a fact about the world. The structure is not.

4. No single thread carries it

Every rate was recomputed 32 times, dropping one thread each time. The placement holds in all 32. The largest movement is AI-loss GRIEF falling from 27.6% to 23.9% when the biggest Character.AI thread is removed, still three times the consumer anchor.

What was predicted, and what happened

prediction, written before any data existed	outcome
anchors separate on GRIEF	confirmed
AI loss above consumer loss, below bereavement, closer to consumer loss	confirmed
AI loss carries consumer-loss levels of RAGE	confirmed , 57.0% vs 56.7%
AI loss shows CARE elevated above consumer loss	failed , it is lower
AI loss shows less SEEKING than consumer loss	withdrawn , the 9.6 point gap is smaller than SEEKING's own 11.8 point instability
instrument instability of 15 to 30%	failed , it is 34%

The SEEKING claim was withdrawn under a rule written before the number existed. That is the only reason the rule is worth anything.

Limitations

- The consumer anchor is one community and one event. r/skype, r/sonos and r/PleX were searched under the same rule and produced no qualifying threads inside the window. The rule was not relaxed to manufacture a corpus.
- The instrument has never been validated against human ground-truth labels. What exists is discrimination, reliability and cross-model replication, which is not the same thing.
- SEEKING is too unstable here to support claims at this precision.
- The largest AI-loss event of the period, the July 2026 shutdown of AI companions in China, is absent: those users write in Chinese and not on Reddit. This is English-language discourse.
- One re-read shard was killed by a content filter on generated output. The trigger was the widowers thread about missing sex, which is the same content carrying the LUST signal. Anyone reproducing this on grief corpora will meet that wall.

Related work

Recent work establishes that this grief is real: the study of the #Keep4o backlash after the GPT-4o retirement, 2026 work on psychologically safe endings for human-AI relationships, and Harvard Business School tracking of Replika users before and after a loss, which found mourning and measurably worse mental health.

This study adds what those cannot: **a scale**. They show that users mourn. This says how much, against two reference points measured with the same instrument, with the noise floor quantified.

Contributions

1. **An empirical report** placing AI-loss discourse between human bereavement and consumer loss, pre-registered, with two failed predictions reported.
2. **A seven-system alternative to valence**, applied at scale, prompts and analysis code released.
3. **A reliability result for LLM-as-annotator affect coding**: 34% within-rater pairwise instability, 39% cross-rater agreement, and corpus structure that survives both.
4. **An answer to the entity question** from the people who lost one: the version, not the relationship, by six to one.